

Wonhyung Choi

M.Sc. Student with 3+ Years of Hands-on Lab Research

Zürich, Switzerland

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Work Experience

Master thesis research

Zürich, Switzerland

University of Zürich, Department of Molecular Life Sciences

Oct 2025 - Present

- Established and developed an affordable ATAC-seq library QC assay by performing qPCR on tagmented material (ATAC-qPCR) with cell-type-specific primers
- Analyzed bulk ATAC-seq data across cell types and nominated cell type specific regions for assay development
- Developed nuclei preparation and Tn5 loading protocols for customized single cell genomics methods

Supervised by Prof. Silvia Domcke, Dr. Giulia Moro and Dr. Erich Brunner

Team HYDROGOAT

Zürich, Switzerland

2025 Bioincubate × Lonza Forge Challenge

Apr 2025 - Oct 2025

- Developed an automated pH feedback control system for 96-well cell culture platforms by installing real-time pH-sensitive sensor on optical fiber with closed-loop pH adjustment.
- Implemented real-time data acquisition and decision making using Raspberry Pi and a Flask-based web GUI to enable an autonomous system.
- Collaborated with ETH D-ITET and the ETH Student Project House Lab.
- Sponsored (CHF 10,000) and supervised by Lonza
- Awarded 3rd Prize (CHF 5,000).

Research assistant

Zürich, Switzerland

University of Zürich, Department of Molecular Life Sciences

Apr 2025 - Aug 2025

- Performed analysis of PBMC sc-Multiome datasets generated on the 10x Genomics and BD Rhapsody platforms to compare data quality and platform-specific performance.
- Optimized bead modification protocols for RoCK and ROI (Robust Capture of Key transcripts and Regions of Interest) using cell capture bead v3 on the BD Rhapsody scRNA-seq platform.

Bachelor thesis research

Beijing, China

Peking University, Key Laboratory of Cell Proliferation and Differentiation

Aug 2023 - Jul 2024

- Investigated the role of transcriptional pausing factors by knock-down subunits of mediator complex during erythropoiesis of human CD34+ hematopoietic stem and progenitor cells.
- Performed in-vitro erythroid differentiation and quantified lineage progression using flow cytometry and transcriptome expression level by qPCR.

Supervised by Prof. Hsiang-Ying Lee and Dr. Shicong Zhu

Research assistant

Beijing, China

Peking University, State Key Laboratory of Membrane Biology

Dec 2022 - Aug 2023

- Investigated the role of a gene during mouse cardiomyogenesis by introducing enhancer and silencer constructs in neonatal mouse cardiomyocyte and HEK-293T cell line.
- Studied transcription factor interactions and its effects on cardiomyocyte gene expression by Co-IP and Western blot.
- Conducted molecular cloning, lipotransfection, and dual-luciferase reporter assays to characterize promoter regulation.

Supervised by Prof. Shiqiang Wang and Dr. Zexin Tong

Education

M.Sc. in Quantitative Biology and Systems Biology

University of Zürich, Department of Molecular Life Sciences

Zürich, Switzerland

Sep 2024 - Dec 2026
(expected)

B.Sc. in Biological Science

Peking University, College of Life Sciences

Beijing, China

Sep 2018 - Jul 2024

Skills

Molecular skills

- Tn5 transposase loading and nuclei preparation for bulk and scATAC sequencing library generation
- Library quality checking using Qubit and TapeStation
- Molecular cloning
- Basic operation, analysis and gating for flow cytometry

Cellular skills

- Mammalian cell culture and differentiation of cell line and primary cell line, including human hematopoietic stem cell and mouse embryonic stem cell
- Nuclei isolation and tagmentation
- Delivering of plasmid and shRNA to cells by lipofection and lentivirus infection
- Immunostaining and imaging for cellular fluorescent image

Computational skills

- Statistical analysis and visualization using R
- Managing large scale data and scheduling jobs via HPC cluster using Slurm
- End-to-end analysis and differential analysis of bulk and scATAC/RNA sequencing data
- Basic knowledge with reproducible workflow (Nextflow)

Documentation

- Managing and producing of reproducible documents using Quarto and Microsoft Office tools
- Version control using Git and GitHub with VSCode

Languages

- Korean (Native), Chinese (C1), English (C1), German (A2)

Publications

MED26-enriched condensates drive erythropoiesis through modulating transcription pausing

Zhu S, Zhang X, Li N, Zhao X, Li M, Xie S, Yue Q, Li Y, Li D, Wu F, Zhang Z, Feng Z, Zhang Y, **Choi W**, Jia X, Deng Y, Hu Q, Yao X, Gao X, Lee HY.

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